

Postsecondary Certificate Trends and Attainment project

Research Progress Report | Week 5

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1 Educational Attainment

The Kentucky panel we built earlier gave us a clean story about spatial inequality in BA attainment inside one state. But a fair question is whether the pattern is specific to Kentucky or whether the state is riding a national wave. The two possibilities have very different implications. If the story is national, then Kentucky’s widening tract-level gap in BA attainment is part of a larger structural shift in where educated adults live in the United States. If the story is Kentucky-specific, then something in state policy, industry mix, or migration is doing the work.

I extended the Kentucky panel to national scope. Every tract in the country from 1990 to 2024 is on 2020 boundaries. The Kentucky numbers replicate the earlier report exactly, so nothing about the state has changed. What is new is that we can now put Kentucky next to the rest of the country and see how much of the widening gap is from Kentucky.

The short answer is that most of it is not. The national tract-level gap in BA attainment has widened just as much as Kentucky’s, and it has widened over the same period. Kentucky is on trend. What sets Kentucky apart is not that the gap is growing, but that it grew from a lower base and is still catching up to the national mean.

2 Data and Panel Construction

2.1 Sources

The national panel uses the same three sources as the Kentucky panel, extended to the full United States. For 1990, I pulled the STF3 educational attainment table for every U.S. tract from NHGIS. For 2000, the equivalent SF3 table, also from NHGIS. For 2012 through 2024, tract-level B15003 estimates from the American Community Survey, pulled through the tidycensus package in

R. Puerto Rico is excluded because 1990 STF3 does not cover it. The other territories are excluded because tract-level ACS coverage for them is inconsistent.

The ACS pipeline pulled 51 state files for each of 13 years, which came to just over one million tract-year rows. The 1990 file has 61,258 tracts nationally. The 2000 file has 65,443. After harmonization to 2020 tract boundaries, all three sources sit on the same structure of roughly 84,400 tracts, matching the Kentucky pipeline structure exactly.

2.2 Harmonization

I used the same rule as the Kentucky panel: crosswalk the head counts first using NHGIS adult-population weights, sum onto the target tract, then compute shares only at the end. Counts add up across the crosswalk. Shares do not.

For the national panel, this meant chaining two crosswalks. The 1990 and 2000 files first go from their native tract boundaries to 2010 tract boundaries, and then from 2010 to 2020 boundaries. The 2012 through 2019 ACS files are on 2010 tracts and only need one crosswalk step. The 2020 through 2024 ACS files are already on 2020 tracts and pass through unchanged.

2.3 Benchmarking

Every year of the panel is benchmarked against a published external total. Table 1 shows the results. National BA attainment shares match NCES Digest and ACS published figures within 0.1 percentage points at every time point. This is the same discipline we applied to the Kentucky panel and it is what gives me confidence the pipeline is producing the right numbers.

Year	Computed BA+ share (%)	Published (%)
1990	20.34	20.3
2000	24.40	24.4
2012	28.5	28.5
2018	31.5	31.5
2024	35.7	35.7

Table 1: National population-weighted BA attainment shares from the harmonized panel, compared to published NCES and ACS totals. All years match within 0.1 pp.

3 Findings

3.1 The national dispersion graph

The main question we asked for Kentucky was whether spatial inequality in BA attainment has widened over time. The answer there was yes, both in the maps and in the P90-P10 gap. Figure 1 runs the same test on the national panel.

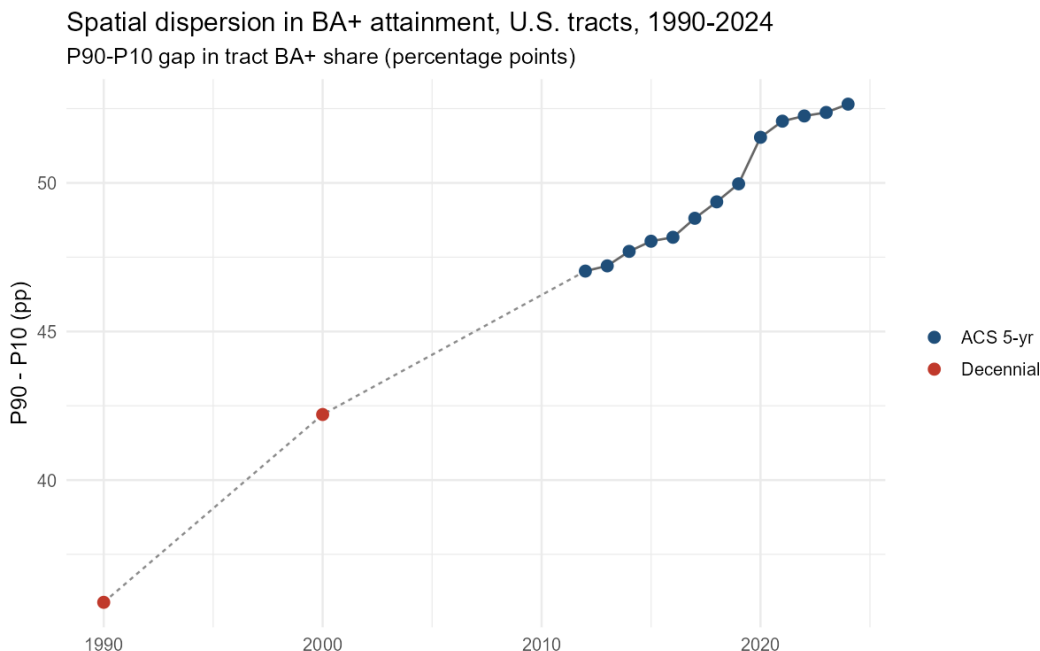


Figure 1: The P90-P10 gap in tract-level BA attainment has widened by nearly 17 percentage points since 1990.

The national gap between the 90th and 10th percentile tracts was 35.9 percentage points in 1990. By 2024 it was 52.7 percentage points. That is a widening of 16.8 percentage points over 34 years. The trend is monotonic. Every ACS year is wider than the one before, and the two decennial years fit smoothly on the same line.

Kentucky's gap widened by 15.8 percentage points over the same period, from 25.4 to 41.2. So the state widened at about the same rate as the country. What is different is the level. Kentucky's gap has always been smaller than the national gap, which makes sense because Kentucky's overall BA attainment is lower and its distribution is compressed toward the bottom. The 90th percentile Kentucky tract in 1990 had a BA share of 29.3 percent, compared to 41.8 percent for the national 90th percentile. Kentucky's high-attainment tracts have never been as high as the country's, and

this is true even in 2024.

The coefficient of variation tells a different story. Nationally, the CV fell from 0.73 in 1990 to 0.59 in 2024. Kentucky's CV fell from 0.87 to 0.64. Both series show relative convergence at the same time as absolute divergence. This is a pattern where the whole distribution shifts up, so the mean rises faster than the spread, and the ratio between them shrinks even as the absolute gap grows.

3.2 The national map

Figure 2 shows BA attainment at the county level for the same four benchmark years used in the Kentucky maps. All four panels use a common color scale, so a color means the same share of adults with a BA in every year. Alaska, Hawaii, and the territories are excluded from the map for readability.

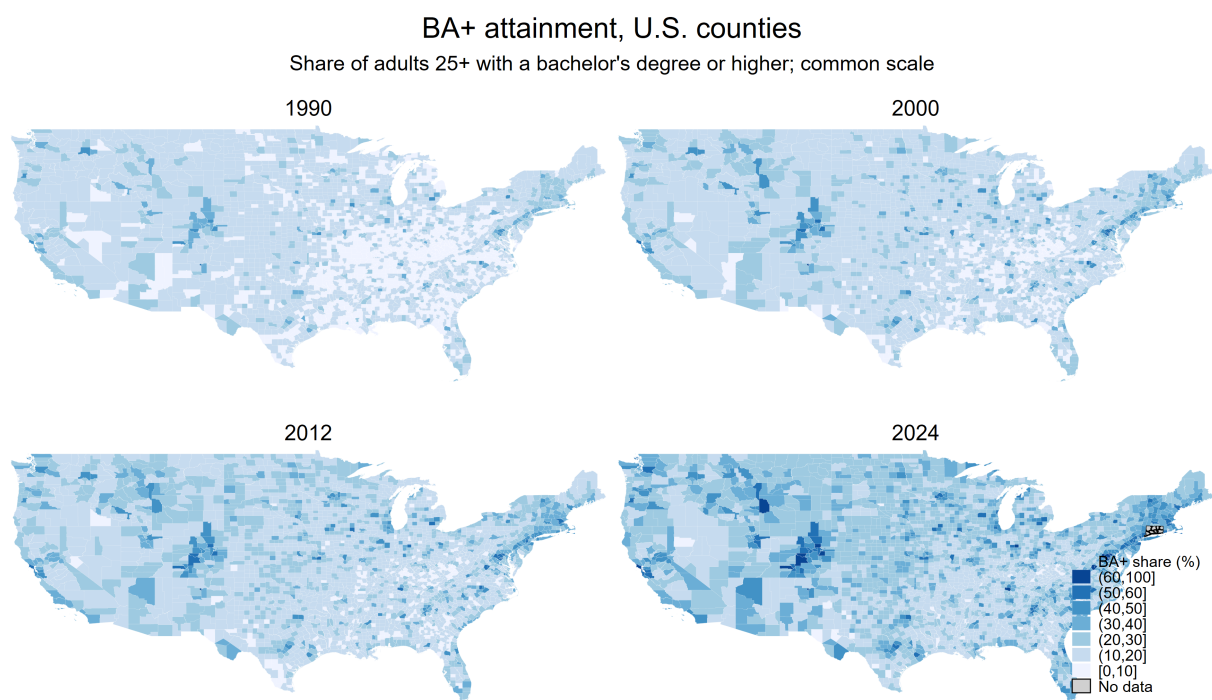


Figure 2: BA attainment by county at four benchmark years. The urban-rural gap deepens over time and clusters intensify in specific regions.

Three patterns show up clearly. First, the map darkens overall. The whole country got more educated over 34 years, and this is the "everyone rose together" component. Second, the darkening

is not uniform. The coasts, the Rockies, and the counties around major research universities pulled ahead at a much faster pace than the rest of the country. By 2024, Bay Area counties, coastal New England, the D.C. metro, Denver-Boulder, Seattle, and the Research Triangle appear as dense dark blue clusters that barely existed in 1990. Third, the low-attainment belt stayed low. Appalachia, the Deep South, the Rio Grande Valley, and parts of the northern Great Plains are still pale in 2024. Kentucky is inside the Appalachian pale zone, which is consistent with the Kentucky-only maps from earlier reports.

3.3 Where Kentucky fits

Rolling the national tract panel up to counties gives a direct way to place Kentucky in the national distribution. Nationally, the mean county grew BA attainment by 11.1 percentage points between 1990 and 2024. Kentucky’s mean county grew by 9.9 percentage points. Kentucky is on the slow side of the national mean, though not dramatically so.

The bottom end of the national county distribution is more striking. Owsley County, Kentucky, has the sixth-slowest BA growth of any county in the United States over the full period. Its BA share went from 9.8 percent in 1990 to 7.1 percent in 2024. It is one of a small handful of counties whose BA share fell over the 34 years. Six Kentucky counties (Owsley, Carroll, Wolfe, Elliott, Lee, Nicholas) sit in the bottom two percent of the national distribution on BA growth. This is a national picture, but the Kentucky story is not incidental to it. The pale corners of the national map that stayed pale for 34 years include most of Eastern Kentucky.

4 Postsecondary Certificate Trend

4.1 Progress this week

The main deliverable this week is a simple time-series chart of credentials awarded nationally from 2000 to 2024, with one line for each of five categories: certificates (all levels combined), associate's, bachelor's, master's, and doctorate. The chart is built from the harmonized IPEDS Completions panel and appears in Figure 3.

Getting the chart to read cleanly across the full period took some care because of a schema break in the IPEDS award codes. Before 2008, doctorates were reported under a single code. Starting in 2008, they were split into three separate codes. The first version of the chart had a flat doctorate line that jumped in 2008, which was a coding artifact. Once the pre-2008 codes were folded back in, the doctorate line runs smoothly across the transition.

Institution counts in the panel are stable and rising slightly from 2020 through 2024, so the bachelor's dip in 2023 and the certificate spike in 2024 appear to be real patterns in the data rather than late reporting.

4.2 The chart

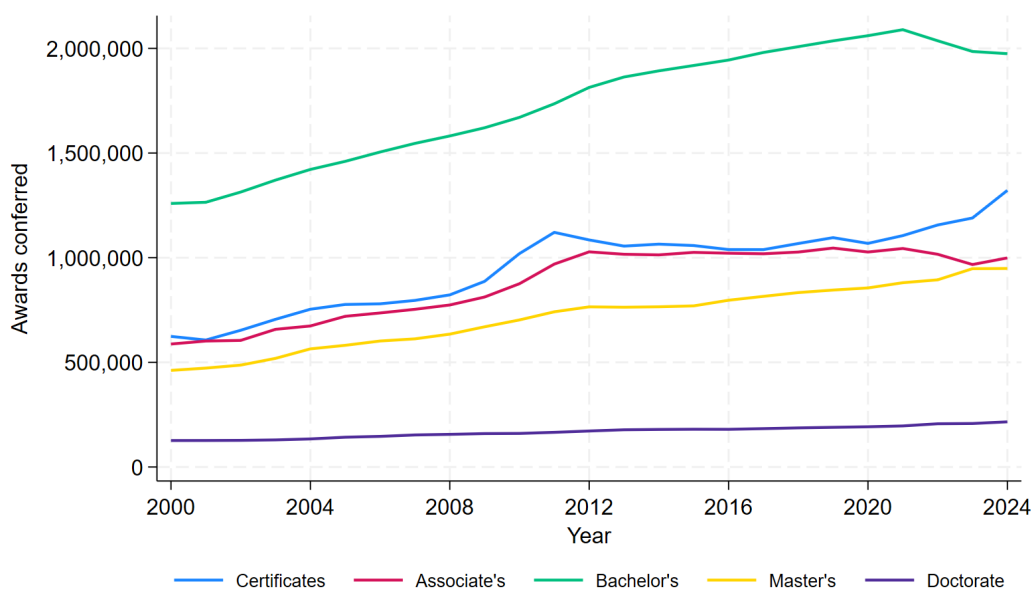


Figure 3: Credentials awarded nationally by category, 2000–2024.

Bachelor's degrees are the largest category throughout the period and grew from about 1.26 million in 2000 to just under 2 million in 2024, with a small dip in the last two years. Certificates and associate's degrees tracked each other closely for most of the 2000s but pulled apart around 2011, and certificates ended the period as the second-largest category. Master's degrees grew steadily and by 2024 have nearly caught up with associate's, which is a notable convergence given they started 130,000 apart in 2000. Doctorates grew slowly and stayed the smallest category.

The certificate line has the most interesting shape. It grew steadily through the 2000s, spiked during the recession, plateaued through the mid-2010s, and then took off again after 2020. By 2024, certificates crossed the 1.3 million mark for the first time in the panel.

4.3 Where things stand

The chart above is the first of two follow-ups Dr. Darolia asked for after reviewing the descriptive report. The second is a rebuild of the panel using predominant award classification instead of the current highest-award classification. I will use it to shape the direction of the analysis for the following weeks according to the guidance I will be provided.